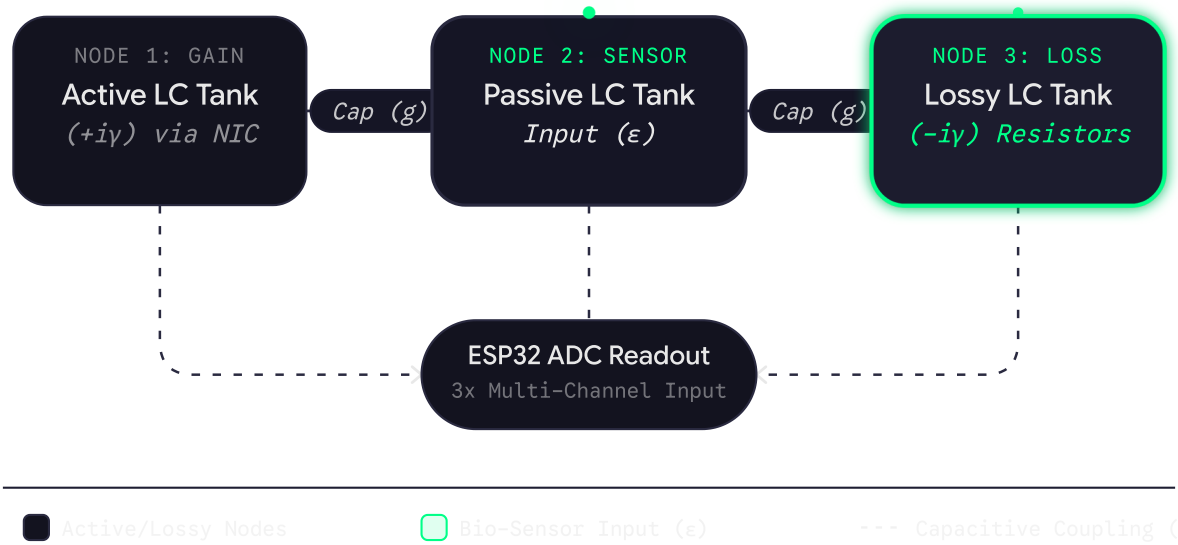
ATLAS • EP3 LAB • #020208 LIVE • 3-NODE

3-Resonator EP3 Circuit Architecture

Exceptional Point 3 (EP3) non-Hermitian sensor topology. Hover/select nodes to view matrix term. Bio-Sensor Input node exhibits cubic-root sensitivity $\Delta\omega \propto \varepsilon^{1/3}$.



Node 3: Lossy LC Tank (Loss)

$H[3,3] = \omega_o - i\gamma$

$$H_{EP3} = \begin{bmatrix} \omega_o + i\gamma & g & 0 \\ g & \omega_o + \varepsilon & g \\ 0 & g & \omega_o - i\gamma \end{bmatrix}$$

Lossy resonator with tailored passive resistors providing dissipation rate -iγ. Completes the gain-neutral-loss EP3 symmetry.

$H[1,1]=\Omega_o+I\Gamma$ • $H[2,2]=\Omega_o+E$ • $H[3,3]=\Omega_o-I\Gamma$ • G = CAP EP3 • ATLAS

EP3 SENSOR — INTERACTIVE VISUAL — BIO-SENSOR INPUT #020208 #e8e8e8 #00ff88